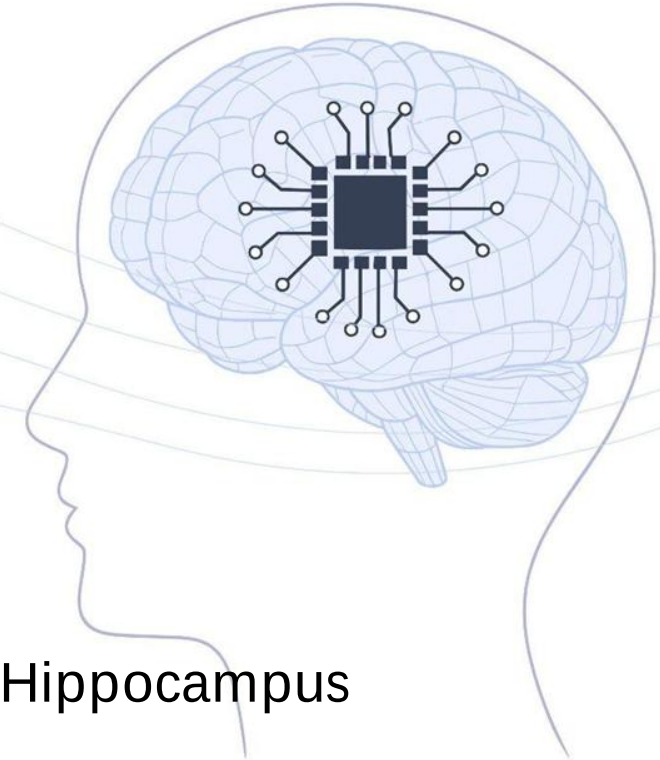
2026 NABI Symposium : Memory In Context

Research Proposal



Memory as Inference, not Relearning

Biological Constraints on Models of the Hippocampus

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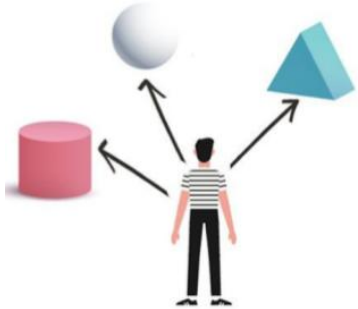
Overview of Today's Proposal

Part 1	Part 2	Part 3	Part 4
Concepts	Motivations & Hypothesis	Prior Research	Methods & Expected Outcomes
◆ Brain and "Cognitive Space" ◆ What does "Biological Plausibility" mean?	◆ The 1D ring task ◆ Zero-shot Inference : Human vs. Memory model	◆ Prior work on SR and TEM ◆ Limitation of Prior work	◆ Methodology : SR and Eigenvectors ◆ Discussion

Concept 1.

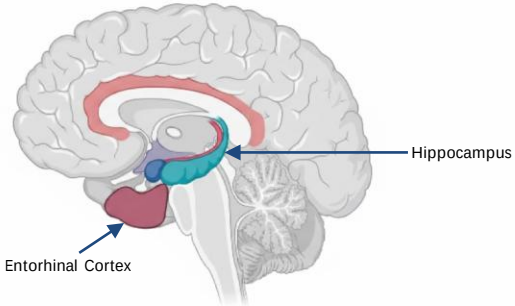


Brain = Cognitive Space

How can humans perceive abstract concepts of space?

1) MEC (Medial Entorhinal Cortex)
 : spacial/geometric coordinates

2) HPC (Hippocampus)
 : spacial navigation, memory formation



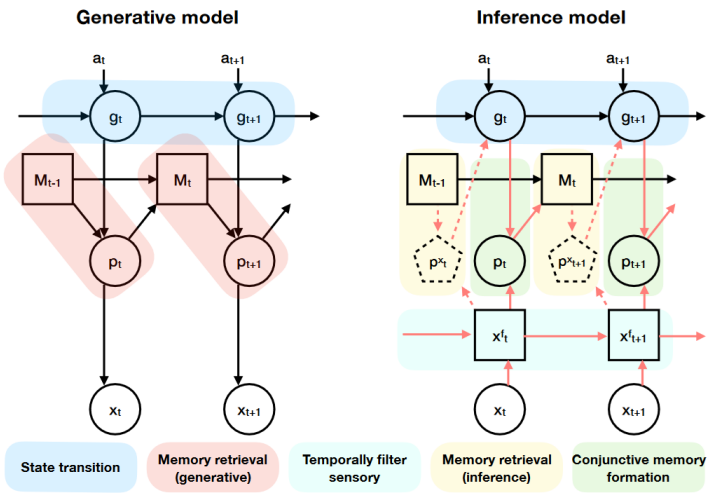
The MEC–HPC system integrates geometric coordinates with memory to construct a cognitive space that enables prediction and inference.

Concept 2.

TEM & SR : Two Computational Views of Hippocampal Memory

1) TEM (Tolman Eichenbaum Machine)

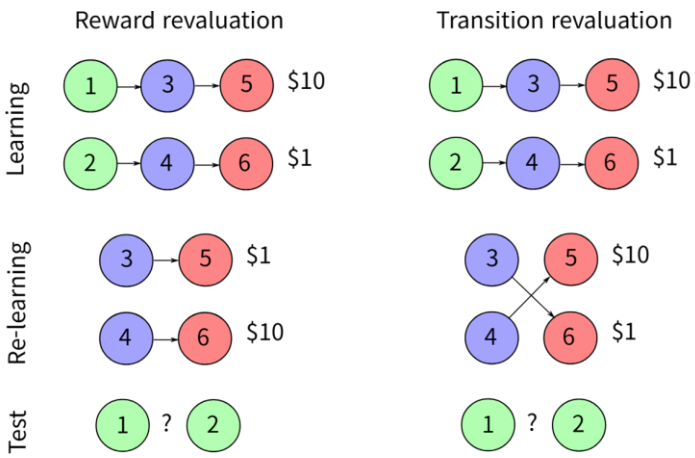
: learns the relational structure of experiences enabling flexible generalization beyond specific locations



Concept 2.

TEM & SR : Two Computational Views of Hippocampal Memory

1) SR (Successor Representation)
: a predictive map, where each location encodes the expected future states that follow from it.



Modeling the MEC-HPC System

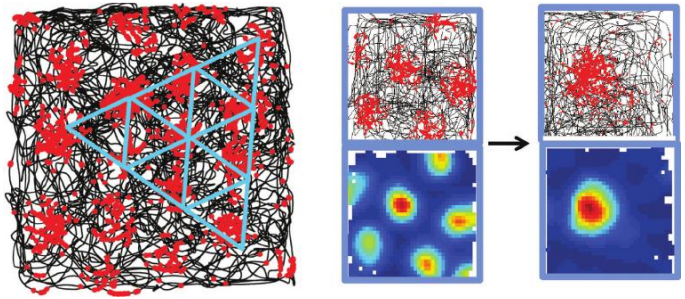
1) Many Computational Models Attempt to Imitate the Hippocampal Formation

- TEM: Explains a Relational Cognitive Map
- SR: Explains a Predictive Map
- Models aim to mimic the function of Grid Cells, Place Cells, etc.

2) Therefore, the "Biological Plausibility" of Models Becomes Important

- Researchers Want to Find a Model that can do:
  - Path Integration and Prediction
  - Zero-Shot Inference
  - Ring Attractor Formation
- A 1D Circular Task is widely used as a Paradigm for Model Learning

Concept 3.



What does "Biological Plausibility" Mean?

1) The Model Learns Like a Human

Ex.1 : TEM models the MEC-HPC system, which makes it a relational cognitive map.

Ex.2 : SR is a predictive map, that associates the HPC with rewards.

2) The Model Shows Results Similar to Human Memory

Ex.1 : TEM produces Place-cell-like Units.

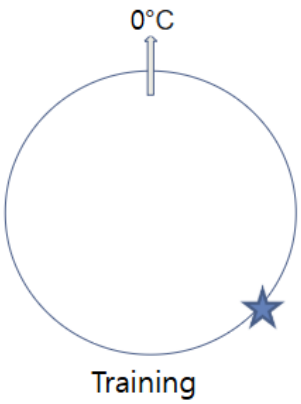
Ex.2 : TEM produces a Grid-Like Structure in Latent Space.

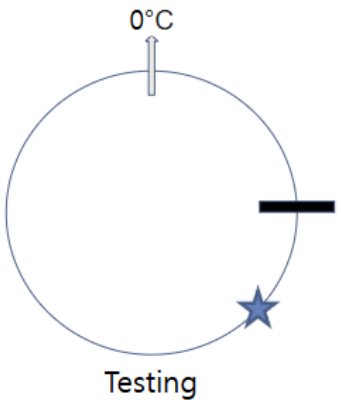
Motivation

A Biologically Plausible Model of the Human Brain should be able to "reuse" existing cognitive maps, not "relearn"

However, under the 1D circular ring framework,
we found differences in the behavior of models and humans.

Task





Task Description

- 1D circular task : hard for neural net to train
  - Primary Cause: Boundary Discontinuity (0≈2π) in Euclidean mapping.
- Train: starting at 0°C open space with reward
- Test: sudden appeared wall at random angle
- Optimal policy:
  Before wall : clockwise
  After wall : clockwise, after meeting wall counter-clockwise

Task

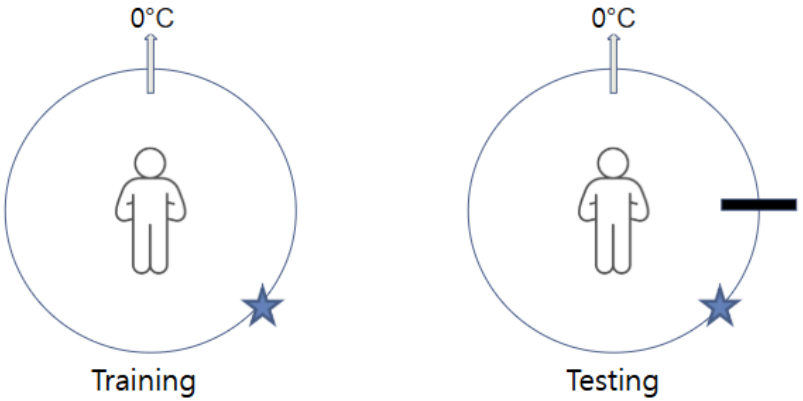
Significance of Task

Model

1D circular task

Human

Head Rotation



In this task, one must understand circle as a angle parameter **θ**, not as a simple linear sequence.

Then, how does human intelligence make judgments in this context compared to the adaptation of an ML agent?

Human vs. ML agent : Judgement Comparison

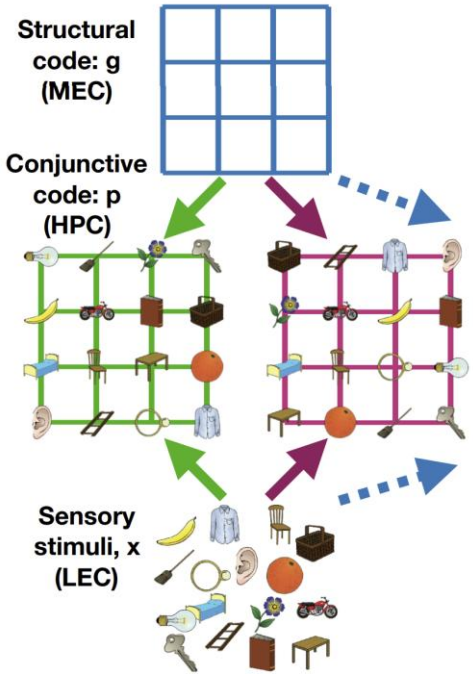
Human		Machine learning agent
◆ The topology	What to understand	◆ The value function or attractor in model
◆ Wall is just obstacle on the pre-learned circle	Perceptual difference between a circle and a wall	◆ Learned circle and wall doesn't have hierarchal difference
◆ Doesn't calculate value of every location again. Vague judgment	Calculation	◆ Calculate global value function to achieve optimal policy

Prior Research

Task learning of TEM

TEM decouples ‘where’ and ‘what’, inferring current state only by memory of pair of action(Path Integration), sensory

However, when meeting obstacle, it needs trials and errors to learn unlearned structure



"NABI-Context in memory"

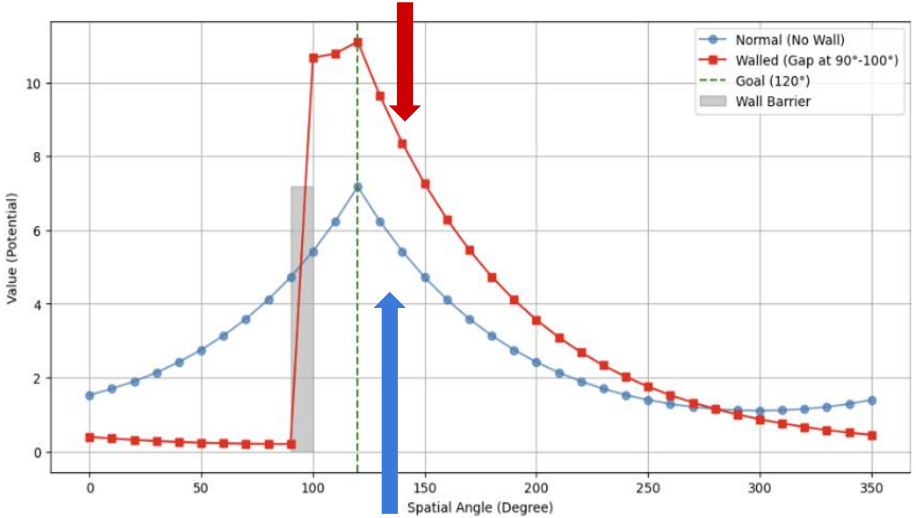
Prior Research

Task learning of SR

SR encodes the expected discounted future occupancy of state along a trajectory from initiated state.

However, in order to reflect a new obstacles, it needs trials and errors to update a predictive map.

Value after wall introduce



Value before wall introduce

$$V = M \cdot R$$

V: value

M: predictive map

R: reward

Prior Research

Limitations of prior research

1) TEM : excels at representing topological structures via continuous attractors □ , but struggles with flexibility in task learning□ .

2) SR : effective for task learning□ based on transition probabilities, yet it fails to achieve zero-shot□ adaptation when the environment is altered.

No model uses learned space, instead focuses on discarding previous information faster.

❋ Zero-shot : perform a task without having seen any specific training data for it beforehand

Proposal

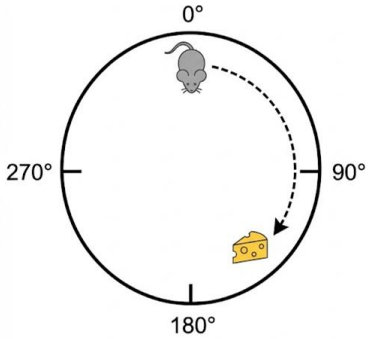
What is Problem?

Existing models must relearn their entire internal map in response to local environmental changes. This fundamentally contradicts our intuitive understanding of the brain.

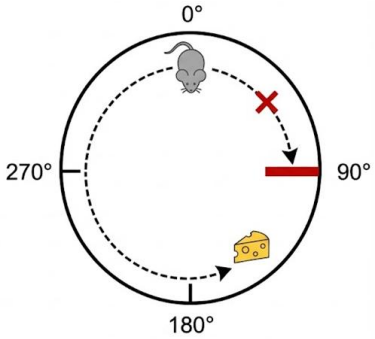
Hypothesis

With an established topological prior, optimal policies against local constraints (like obstacles) can be inferred by recombining original eigenvectors, bypassing the need for global map retraining.

Methodology



M



M'

$$V = M \cdot R$$

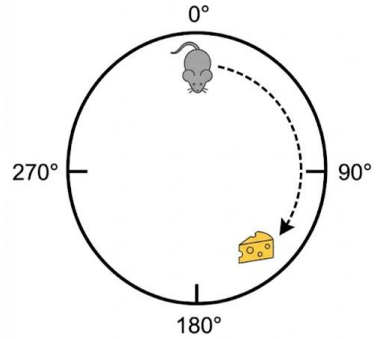
V: value; M: predictive map; R: reward

How can we make this M'?

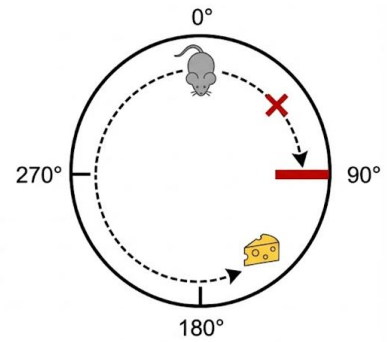
SR updates M' through trial and error by exploring the space.

It becomes equivalent to recalculating the entire spatial network from scratch.
(i.e. computing the inverse matrix)

Methodology



M



M'

$$V = M \cdot R$$

V: value; M: predictive map; R: reward

How can we make this M'?

Can an agent adapt to changes in the external environment M' by leveraging its existing knowledge M, rather than relying on trial and error?

need for hierarchical change for local change

Methodology

M

predictive map

↕

Eigen vectors

Grid cell

According to Stachenfeld, it is known that eigenvectors of M represent grid cells. M can be approximated from its eigenvectors, grid cells, and by only adjusting their weights,

we assume that the changed environment M' can be approximated by grid cells using Spectral Filter, leaving the original grid cells perfectly intact.

❋ Spectral Filter: A mathematical transformation that selectively filters specific information by decomposing data via an eigenbasis and adjusting its eigenvalues

Discussion

1. Task analogy and generalization

1D circle task

Function	MEC layer
Head rotation	Head direction cell
Body Rotation	Grid cell

The existence of a prior basis may be leveraged for effective task-solving

Discussion

2. Biological plausibility

1. Continuous Input of Prior Eigenvectors

- Neural Basis: Supported by neural circuits that maintain and provide a prior eigenvectors

utilizing pre-existing eigenvectors for spatial navigation.

2. Timescale Separation: Spatial Representation vs. Policy Learning

- Speed Discrepancy: Finding an optimal policy using *prior* eigenvectors is significantly faster than the physical updating of grid cells driven by boundary changes.

achieve immediate adaptation by leveraging existing knowledge, rather than reconstructing the spatial map.

3. Cognitive Asymmetry in Boundary Appearance vs. Disappearance

- Learning Dynamics: Suggests biological and algorithmic asymmetries in how the agent learns when a new boundary *appears* versus when it *disappears*

inherently comprehends the 'prior' or original structure of the environment.

Discussion

3. Expected Results

Phase-Integrated Model:

Integrate physical grid cell updates (spatial recognition) and fast policy updates via existing grid cells (initial filtering).

Prior-Based RL Methodology:

Develope a new RL approach that uses prior spatial representations (eigenvectors) to rapidly adapt to environmental shifts.

Prior Dependency Parameter:

Analyze behavioral differences by modeling how much different tasks rely on pre-existing spatial knowledge (e.g., Rotation vs. 1D circle)

Thank you for listening